URRT

URC

URRT ROBO CUP

GENERAL RULEBOOK

01

MOBILITY

Rails slopes gravel and narrow paths

02

DEXTERITY

QR codes touch targets box smash and the sensor crate

03

AUTONOMY

Hands-off execution worth far more points

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CONTENTS

1 Introduction . . . 3
- 1.1 Purpose of the Championship . . . 3
- 1.2 The Three Pillars . . . 3
- 1.3 How to Read This Rulebook . . . 4

2 Definitions . . . 5

3 Championship Overview . . . 6
- 3.1 Format at a Glance . . . 6
- 3.2 Participating Teams and Run Allocation . . . 6
- 3.3 The Lane Unlock Rule . . . 7
- 3.4 Advancement . . . 7

4 Team Development Milestones . . . 7
- 4.1 Milestone Schedule . . . 7
- 4.2 Milestone Rules . . . 8

5 The Arena . . . 9
- 5.1 Arena Overview . . . 9
- 5.2 Walls . . . 11
- 5.3 Lane Overview . . . 11
- 5.4 Lane 1 – X Rail . . . 11
- 5.5 Lane 2 – 30° Slope with X-Rail and Gravel . . . 12
- 5.6 Lane 3 – K-Rail and Narrow Paths . . . 12
- 5.7 Operator Station . . . 12
- 5.8 Arena Services and Officials . . . 12

6 Mobility Pillar . . . 13
- 6.1 Continuous Traverse . . . 13
- 6.2 Mobility-Only Trials . . . 13

7 Dexterity Pillar . . . 13
- 7.1 Dexterity Tasks . . . 13
- 7.2 Box Smash – Lane 1 . . . 14
- 7.3 Optional Tasks . . . 14

8 Scoring . . . 14
- 8.1 Scoring Formula . . . 14
- 8.2 Base Points . . . 15
- 8.3 Multiplier Summary . . . 15
- 8.4 Worked Example – QR Decode and Ping . . . 15
- 8.5 Build Quality Score . . . 16
- 8.6 Rounding and Verification . . . 17

9 Run Procedure . . . 17
- 9.1 Before the Run . . . 17
- 9.2 During the Run . . . 17
- 9.3 After the Run . . . 18

10 Penalties and Disqualification 18
10.1 Penalty Schedule 18

11 Team Conduct and Fair Play 19
11.1 Conduct Rules 19

12 Ranking and Awards 19
12.1 Final Ranking 19
12.2 Awards 20

13 Protests and Rule Interpretation 20
13.1 Protest Procedure 20

14 Revision History 21

DOCUMENT SET

This is the **General Rulebook**, part of the four-document URRT Robo Cup set:

- **General Rulebook** – format, milestones, arena, tasks, scoring including the Build Quality Score, run procedure, penalties, ranking.
- **Mechanical & Electrical Rulebook** – rover size, mechanical design and CAD, electrical and wiring rules, E-stop, power, communications, classes, safety.
- **Software & Autonomous Rulebook** – control modes, autonomy, sensor crate, dashboard.
- **Judgement Sheet** – inspection, score, dashboard and build quality sheets, milestone form, judge quick reference.

Where a rule lives in another document, this book points to it by name.

1. INTRODUCTION

1.1 Purpose of the Championship

The URRT Robo Cup (URC) tests rescue rovers built by interns on **reliability** across a set of standardized lanes. The event is modeled on the RoboCup Rescue Robot League: robots are not judged on how impressive they look, but on how consistently they can traverse difficult terrain, interact with objects, gather sensor information and, when the team dares, do it all without a human operator.

The standardized lanes allow results to be compared fairly between teams and between seasons. Every rule in this book exists to serve one of three goals:

- to reward rovers that work repeatedly and safely,
- to reward teams that take the risk of higher autonomy, and
- to keep participants, officials, spectators and the arena safe.

1.2 The Three Pillars

The competition is built on three core pillars. Every task in the arena belongs to at least one of them.

Pillar	What is tested	How it is scored
Mobility	Terrain navigation: rails, slopes, gravel and narrow paths.	Continuous end-to-end traverses of a lane.
Dexterity	Object interaction and sensor tasks: inspecting QR codes, touching targets, smashing boxes, using the sensor crate.	Points per completed task.
Autonomy	Hands-off execution of any eligible task.	Multiplier on the task score (up to $\times$**4**); worth significantly more than teleoperation.

Table 1. The three pillars of the URRT Championship.

1.3 How to Read This Rulebook

1.3.1 The words **must**, **shall** and **is not permitted** describe mandatory requirements. **Should** and **may** describe recommendations and options.

1.3.2 Rules are numbered by section so that referees, teams and protests can refer to them precisely (e.g. “Rule 5.2.3”).

1.3.3 Dimensions marked *nominal* may vary by $\pm 10\%$ in the arena; teams must design with margin.

1.3.4 In case of conflict, the order of precedence is: safety rules, then these written rules, then the Head Referee’s decision on the day.

2. DEFINITIONS

Term	Definition
Arena	The whole competition area, containing Lanes 1–3, the start zones and the operator station.
Lane	A standardized test section with its own terrain, dexterity tasks and optional tasks.
Run	One timed attempt by one team in one stage. A run may cover several lanes.
Operator	The single team member who commands the rover from the operator station.
Operator Station	The area, screened from the arena, from which the operator works.
Milestone	A scheduled progress checkpoint at which a team reports and demonstrates the state of its rover to the organizing committee (Section 4).
Dashboard	The team-built web interface, running on the rover's own circuit and network, used to monitor, switch modes and control the rover (the Software & Autonomous Rulebook).
Dashboard Score	The marks (maximum 100, plus an optional bonus) earned in the Dashboard Review and added to the team's final total.
Build Quality Score	The marks (maximum 30) earned at the M4 Build Review for soldered circuits, power connectors, wire colours, circuit schematics and CAD design files; added to the team's final total.
Teleoperation (Teleop)	Control in which a human issues commands that directly or semi-directly drive the rover.
Autonomous Mode	Control in which the rover decides and acts on its own after the start command; no control input from a human is permitted.
Traverse	A continuous, end-to-end passage of a lane from the start line to the finish line.
Reset	Returning the rover to the lane start by hand after a failure or at the team's request.
Intervention	Any human action on the rover during a run (reset, repositioning, touching, switching mode).
Sensor Crate	The standardized set of sensing tasks (audio, QR, sonar, ping, colour/shape, motion) described in the Software & Autonomous Rulebook.
Base Points	Points assigned to a task in the arena before any multiplier is applied.
Class Multiplier	Multiplier set by the mobility configuration of the rover (4WD, 2 flippers or 4 flippers).
Control Multiplier	Multiplier set by the control mode used for a task (teleop or autonomous).
Head Referee	The final on-site authority on interpretation of the rules.

Table 2. Key terms.

3. CHAMPIONSHIP OVERVIEW

3.1 Format at a Glance

The championship runs in three stages. Each stage uses the same lanes, but the content and the lanes available to a team change as it progresses.

Stage	Name	Content	Purpose
1	Qualification Trials	All three lanes; mobility and dexterity tasks; teleop or autonomous.	Establish a baseline score and unlock lanes.
2	Mobility-Only Trials	Lane traverses only; no dexterity or sensor tasks.	Prove terrain reliability and unlock lanes.
3	Semi-Finals and Finals	Full sequences on unlocked lanes only.	Determine the champion.

Table 3. Championship stages.

3.2 Participating Teams and Run Allocation

Four teams take part in the championship. Each team enters exactly one rover. The run order inside every stage is drawn by the organizers at the technical briefing, and the order is reversed for the final so that no team always runs first.

Stage	Teams running	Runs per team	Run length
1 – Qualification	All four teams	2	10 min
2 – Mobility-only	All four teams	2	8 min
3 – Semi-final	Teams with at least one unlocked lane	1	10 min
3 – Final	Top two teams of the semi-final	1	10 min

Table 4. Run allocation for four teams.

3.3 The Lane Unlock Rule

UNLOCK RULE

A team must score **at least 3 points** in a lane during the first two stages (Stage 1 and Stage 2) to **unlock** that lane for Stage 3. Lanes that are not unlocked are closed to the team in the semi-finals and finals.

3.3.1 Points counted toward the unlock threshold are *base points* (before class and control multipliers), so that a simple 4WD rover is not disadvantaged against a flipper rover.

3.3.2 Points from Stage 1 and Stage 2 in the same lane may be combined to reach the threshold.

3.3.3 A team that unlocks no lane is not eligible for Stage 3 but keeps its Stage 1 and 2 results.

3.3.4 The unlock status of every team is published after Stage 2 and cannot be appealed on the grounds of score recalculation more than 30 minutes after publication.

3.4 Advancement

3.4.1 Teams are ranked after Stage 2 by the sum of their Stage 1 and Stage 2 scores.

3.4.2 All teams that have unlocked at least one lane play the semi-final. The top two teams of the semi-final advance to the final; the remaining teams are ranked by their semi-final result for third place and below. Organizers may change this and will announce any change at the team briefing.

3.4.3 Stage 3 scores are added to the earlier scores only for the final ranking as defined in Section 12.

4. TEAM DEVELOPMENT MILESTONES

4.1 Milestone Schedule

Each of the four teams is guided through the build by **four milestone-based updates**. At every milestone a team submits a one-page written update (the Judgement Sheet), gives a ten-minute demonstration to the organizing committee and answers questions. The milestones follow the three pillars and finish with a full readiness review before the championship.

Milestone	Timing*	Deliverables	Pass criteria
M1 – Design Review	1 week	Written design: class declaration (A, B or C), CAD model within $300 \times 300 \times 300$ mm, circuit schematic draft (simulated or hand-drawn, pins labelled), E-stop concept, plan for the QR and ping task, GitHub repository created, dashboard wireframe.	Design meets the Mechanical & Electrical Rulebook; class declared; E-stop concept is hardware-based.
M2 – Mobility Demo	1 week	Working rover drives on a practice rail section; E-stop demonstrated; short video of a Lane 1 style crossing.	Rover crosses an X-rail pair under control; E-stop cuts all actuator power.
M3 – Dexterity and Sensor Demo	1 week	Live demo of QR decode and ping in teleop, touch target with the chassis or flipper, and a box smash test; dashboard shows live sensor data and camera.	Decode and ping logged by the organizers' server; at least one box dislodged; at least one other sensor crate task shown; dashboard polls live data from the rover circuit.
M4 – Autonomy and Readiness Review	1 week	Autonomous QR decode and ping demo, full technical inspection, a timed dry run in the practice arena, the Build Review (schematic and CAD files) and the Dashboard Review (the Software & Autonomous Rulebook).	Passes inspection; autonomous task declared and demonstrated or formally withdrawn; dry run completed within the time limit; Dashboard Review and Build Review done; repository snapshot with schematic and CAD files submitted.

Table 5. The four development milestones. *T is the first day of the championship; timings are nominal.

4.2 Milestone Rules

4.2.1 Every team must complete all four milestones. Updates are submitted to the Team mentor at least 24 hours before the milestone session.

4.2.2 Each milestone is given one of three statuses: **Passed**, **Conditional** (pass with listed actions to be completed before the next milestone) or **Not passed**.

4.2.3 A team that does not pass M4 is not permitted to run in Stage 1 until it passes technical inspection and a dry run with a referee.

4.2.4 Milestones are not scored and do not add championship points. Milestone results feed into the Best Engineering award (Section 12).

4.2.5 All four milestone sessions are open to the other teams so that teams can learn from each other; confidential design details may be withheld at the team's request.

4.2.6 Dates may only be moved by the organizing committee and only for all four teams at once.

Status	Meaning
Passed	All pass criteria met. No action required.
Conditional	One or more pass criteria are missing; the committee lists the actions and a deadline, usually the next milestone.
Not passed	The main pass criteria are missing; the team repeats the milestone within 2 days.

Table 6. Milestone status levels.

5. THE ARENA

5.1 Arena Overview

The arena is a closed, fixed-layout test area with three standardized lanes side by side. Every lane has its own start zone, terrain section, dexterity station and finish line. Lanes are separated by walls, so a rover can never see or touch an adjacent lane. The operator station is placed outside the arena behind a visual barrier, so that the operator can only use the rover's own sensors to see.

Item	Specification (nominal)
Arena footprint	Approximately 4.0×5.5 m for the three lanes, start zones and finish zones.
Number of lanes	Three (Lane 1, Lane 2, Lane 3), side by side.
Lane width	1.2 m; narrow sections in Lane 3 are 450 mm.
Lane length	Lane 1: 4.0 m; Lane 2: 4.0 m measured along the slope; Lane 3: 4.5 m.
Wall height	**More than 300 mm** everywhere (nominal 350 mm), see Section 5.2.
Wall finish	Rigid, smooth, matt dark panels; no reflective surfaces.
Floor	Lane 1 and 3: sealed plywood or equivalent with rails fixed on top. Lane 2: sloped plywood with a loose gravel bed.
Start zone	0.5 m deep $\times$ 1.2 m wide at the bottom of each lane, with a marked start line.
Finish line	Marked line at the end of each lane; the dexterity station lies in the last metre before it.
Lighting	Uniform indoor lighting, 300–600 lux, no direct sunlight on the lanes.
Operator station	At least 3 m from the arena and behind an opaque barrier.
Network	One organizer-provided access point and one referee ping server for the arena.
Tolerance	$\pm 10\%$ on all nominal dimensions, except wall height which is never below 300 mm.

Table 7. Arena specification.

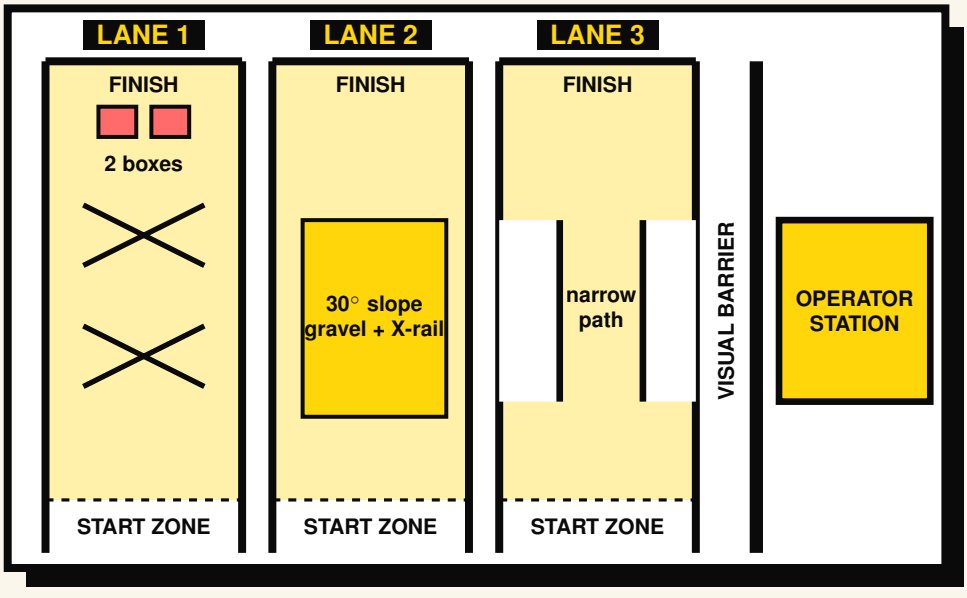


Figure 1. Arena layout. The operator station (right) is outside the arena behind a visual barrier.

5.2 Walls

5.2.1 All lane walls, including the walls between lanes and the end walls, are **taller than 300 mm** (nominal 350 mm). On the slope in Lane 2 the height is measured perpendicular to the local floor.

5.2.2 Because walls are taller than the maximum rover height of 300 mm, a rover cannot climb out of a lane and cannot use another lane or the outside area as part of its path.

5.2.3 Walls are rigid and smooth. Rovers may touch or push against walls, but climbing a wall or leaving the lane over it is treated as leaving the lane (Section 10).

5.2.4 In Lane 1 within the wall there exists two boxes on its top edge (Section 7.2). The rover reaches them by impact on the wall, not by climbing (i.e. the rover has to smash into the wall to displace the box).

5.2.5 The K-rail elements of Lane 3 are integrated within the walls (Section 5).

5.3 Lane Overview

	Lane 1	Lane 2	Lane 3
Terrain	Rail × rail (X-rail) hurdles	30° slope with X-rail and gravel	K-rail (rails within the wall) and narrow paths
Main challenge	Climbing and crossing rail pairs	Climbing with loose gravel traction	Precision and clearance in confined spaces
Dexterity station	Yes	Yes	Yes
Special element	Two wall-edge boxes to smash	Gravel bed on the slope	Narrow path sections
Difficulty	Entry	Intermediate	Advanced

Table 8. Lane overview.

5.4 Lane 1 – X Rail

5.4.1 Lane 1 is a nominal 1.2 m* wide and 2 m* long corridor with crossed rails (X-rails) placed in series on the floor. The rails are timber or equivalent, nominal 50×100 mm*.

5.4.2 The rails are placed so that the rover must cross a pair of rails in the form of an X before continuing. Rail spacing is not standardized beyond the nominal values above.

5.4.3 Somewhere within the lane, at the edge of the wall, two boxes are placed on the wall edge (the *box targets*). The rover must smash into the wall to displace the boxes (see Section 7.2).

5.4.4 The Lane 1 dexterity station is located before the box targets.

5.5 Lane 2 – 30° Slope with X-Rail and Gravel

5.5.1 Lane 2 is a nominal 1.2 m* wide lane built around a 30° slope surfaced with a bed of loose gravel (nominal 20–40 mm* crushed stone).

5.5.2 X-rails are mounted on the slope, so the rover must climb while crossing obstacles on loose ground.

5.5.3 The Lane 2 dexterity station is placed at the top of the slope.

5.5.4 Gravel is raked flat by the officials before every run.

5.6 Lane 3 – K-Rail and Narrow Paths

5.6.1 Lane 3 includes K-rail sections, which are rails integrated within the lane walls, and sections where the path narrows to a nominal 450 mm* width.

5.6.2 The narrow paths are intended to test accurate steering, clearance of the flippers and body, and sensing in confined space.

5.6.3 The Lane 3 dexterity station is placed after the narrow section.

*subject to change

5.7 Operator Station

5.7.1 The operator station is visually shielded from the arena. The operator may only use the rover's own cameras and sensors to understand the situation.

5.7.2 Only one operator and one tether handler (if applicable) are allowed in the operator station during a run. Other team members may observe from the team area.

5.7.3 The team may not communicate with the operator through any channel during an autonomous task.

5.8 Arena Services and Officials

5.8.1 One referee is assigned to each lane, and a Head Referee supervises the whole arena. A timekeeper runs the official clock, which is visible to the operator.

5.8.2 Each run is recorded on video for score verification and protests.

5.8.3 The organizers' ping server records every echo request with a time stamp and the source address, and its log is the official proof for the ping task.

5.8.4 Between runs, officials reset the boxes, rake the gravel in Lane 2, replace QR panels and check that all lane elements are in their marked positions.

5.8.5 Teams get a practice slot in the arena before Stage 1 and may measure the lane elements, but may not alter them.

6. MOBILITY PILLAR

6.1 Continuous Traverse

6.1.1 A **traverse** is a continuous, end-to-end passage of a lane from the start line to the finish line without a reset, a human intervention or leaving the lane.

6.1.2 The traverse starts when the entire rover is behind the start line and the referee gives the start command. It is complete when the rover's centre has fully passed the finish line.

6.1.3 A rover that falls over but recovers by itself without intervention may continue the traverse.

6.1.4 A rover may traverse a lane several times during a run; each completed traverse scores again, up to the traverse limit stated in Table 9.

6.2 Mobility-Only Trials

6.2.1 In Stage 2 only mobility is scored. Dexterity and sensor tasks give no points, and boxes in Lane 1 are displaced.

6.2.2 The control multiplier and class multiplier apply to the traverse score as normal.

7. DEXTERITY PILLAR

7.1 Dexterity Tasks

Every lane has a dexterity station. Dexterity tasks require the rover to interact with an object using its chassis, flippers or on-board sensors. Rovers have no arm or gripper.

7.1.1 **Inspect QR code:** the rover positions its camera so that the QR code can be decoded (the Software & Autonomous Rulebook).

7.1.2 **Touch target:** the rover touches a marked target plate with its front bumper, chassis or a flipper. Contact must be sustained for 2 seconds and detected by the referee.

7.1.3 **Box smash (Lane 1 only):** see Section 7.2.

7.1.4 The rover must not damage the lane, the QR panel or the target plates. Intentional damage leads to penalties (Section 10).

7.2 Box Smash – Lane 1

Two boxes are placed at the edge of the wall at Lane 1. The rover must smash into the wall to eliminate them, and the team earns points based on the displacement of boxes from their home position.

7.2.1 A box is **eliminated** when it is knocked off the wall edge and lands on the arena floor by the rover's impact (Full Displacement).

7.2.2 The rover shall smash the boxes by driving into the wall or by using its flippers. Throwing, firing or releasing any part of the rover is not advisable (but permitted).

7.2.3 The impact must not cause any detachment of rover parts. A rover that loses parts in the arena must be checked by the Technical Committee before the next run.

7.2.4 The boxes are standardized, lightweight and refilled between runs. Boxes that are displaced by anything other than the rover are not counted.

7.2.5 Points are awarded based on displacement per cm (Table 9). Displacing both boxes fully in one run earns an additional completion bonus.

7.3 Optional Tasks

Optional tasks are available in the lanes and give extra points, but are never required for a lane to be unlocked.

7.3.1 **Payload carry:** a referee places a standard 500 g payload on the rover's on-board carrier tray at the start line. The rover carries it through the lane and delivers it into the marked drop zone, by tilting the tray or using its flippers, without losing it on the way.

7.3.2 **Leaning obstacle:** the rover negotiates a leaning obstacle (a leaning panel, pallet or board) by climbing over it or passing underneath it without moving it out of its marked position.

7.3.3 Optional tasks are scored with the same class and control multipliers.

8. SCORING

8.1 Scoring Formula

TASK SCORE FORMULA

$$\text{Task score} = \text{Base points} \times \text{Class multiplier} \times \text{Control multiplier}$$

The total run score is the sum of all task scores in the run, minus penalties.

8.2 Base Points

Lane	Task	Base points	Max per run
	Continuous traverse	4	2 traverses
	QR decode / ping	3 + 3	1
	Touch target	3	1
	Box smash (per box & per cm)	5*per cm displacement	2 boxes
	Both boxes bonus	2	1
	Continuous traverse	6	2 traverses
	QR decode / ping	3 + 3	1
	Colour/shape detection	3	1
	Two-way audio acuity	2 + 2	1
	Optional: payload carry	4	1
	Continuous traverse	8	2 traverses
	QR decode / ping	3 + 3	1
	Motion tracking	4	1
	Sonar proximity	3	1
	Optional: leaning obstacle	4	1

Table 9. Base points per lane and task.

8.3 Multiplier Summary

Control mode	Class A (4WD)	Class B (2 flippers)	Class C (4 flippers)
Teleoperation (×1)	×1	×2	×4
Autonomous (×4)	×4	×8	×16

Table 10. Combined multiplier = class multiplier × control multiplier.

8.4 Worked Example – QR Decode and Ping

The QR decode and ping task is worth $3 + 3 = 6$ base points. The table shows what the same task is worth for every combination of class and control mode.

Rover class and mode	Base	Combined mult.	Task score
Class A, teleop	6	×1	6
Class A, autonomous	6	×4	24
Class B, teleop	6	×2	12
Class B, autonomous	6	×8	48
Class C, teleop	6	×4	24
Class C, autonomous	6	×16	96

Table 11. QR decode and ping scores by class and control mode.

EXAMPLE RUN

A Class B rover (×2) enters Lane 1. It completes one traverse by teleop ($4 \times 2 \times 1 = 8$), decodes the QR code and pings the address autonomously ($6 \times 2 \times 4 = 48$), and eliminates both boxes by teleop ($(5 + 5 + 2) \times 2 \times 1 = 24$). The run total is **80 points**. Its base points in the lane are $4 + 6 + 12 = 22$, well above the 3-point unlock threshold.

8.5 Build Quality Score

Good build practice earns points. At the M4 **Build Review** the reviewers tick each item below against the rover and the repository. The requirements themselves are in the Mechanical & Electrical Rulebook.

Check	Item	What the reviewer checks	Max marks
☐	**Soldered circuit**	All circuits on soldered veroboard or PCB; no breadboard anywhere on the rover.	5
☐	**Power connectors**	XT or T connectors on all power connections; no bare wire, clips or taped joints.	3
☐	**Wire colours**	Red for every positive power line, black for GND, other colours for signals.	3
☐	**Circuit schematic**	Made in circuit software or neat by hand; every pin mentioned; power rails and voltages shown.	6
☐	**CAD design files**	Rover designed in CAD; native files and STEP/STL exports in the repository.	6
☐	**3D-printed chassis**	Chassis (main structure) is 3D printed; printable files available.	3
☐	**CAD matches rover**	Built rover matches the CAD model and the schematic matches the circuit.	2
☐	**Files in repository**	Schematic, pin map and CAD files are in the GitHub repository before the freeze.	2
Total			**30**

Table 12. Build Quality marking scheme.

8.5.1 The Build Review takes place with the technical inspection at Milestone M4 and lasts 10 minutes per team. The repository is frozen at the M4 deadline and files added later earn no marks.

8.5.2 A checkmark earns its marks only if the item is compliant when first reviewed. A mandatory item that is corrected afterwards lets the rover pass inspection and run, but earns no marks.

8.5.3 The circuit schematic and the CAD design files may earn half marks if they are incomplete (for example pins missing, or only part of the rover modelled).

8.5.4 The **Build Quality Score** is the sum of all marks in Table 12 and is added once to the team's final total.

8.6 Rounding and Verification

8.6.1 All scores are integers; fractions are not used.

8.6.2 Referees record every scored task on the score sheet (the Judgement Sheet) and the team captain signs the sheet at the end of the run.

8.6.3 Disputes about a score must be raised before the team captain signs the sheet.

9. RUN PROCEDURE

9.1 Before the Run

9.1.1 Teams are called to the staging area five minutes before the scheduled start. A team that is not ready 5 minutes after its call forfeits that run.

9.1.2 Teams have 3 minutes in the arena to set up the rover at the start line and to check communications.

9.1.3 The team declares to the referee which lanes it will attempt and which tasks it will attempt autonomously.

9.2 During the Run

9.2.1 A run lasts a maximum of 10 minutes in Stage 1 and Stage 3 and 8 minutes in Stage 2. The clock starts with the referee's start command and is not stopped for repairs.

9.2.2 Only the operator and the referee are allowed to interact with the rover or the operator station during a run.

9.2.3 The rover may switch lanes in the order the team chooses, but Stage 3 allows only unlocked lanes.

9.2.4 Reset and intervention rules are described in Section 10.

9.3 After the Run

9.3.1 The rover shall be powered off or have its E-stop pressed before anyone touches it.

9.3.2 The referee reads out the score, and the captain signs the score sheet.

9.3.3 The team clears the arena within 2 minutes.

10. PENALTIES AND DISQUALIFICATION

10.1 Penalty Schedule

Infraction	Penalty	Notes
Reset (rover returned to lane start by hand)	Current traverse void	Time continues to run; points already scored are kept.
Third reset in the same lane	Lane closed for the run	The team may move to another lane.
Leaving the lane or hitting a wall hard	-2 points	Judged by the referee.
Damage to a lane element or QR panel	-5 points	Per incident; repairs delay the next team.
Operator input during a declared autonomous task	Task becomes teleop ($\times 1$)	If the input is a control command; see the Software & Autonomous Rulebook.
Ping sent from off-board device	Task void	Zero points for the ping.
E-stop failure at inspection or during the run	Run stopped	Must be repaired and re-inspected.
Unsafe behaviour or ignoring a referee	Disqualification from the run	The Head Referee may extend it to the stage or championship.
Intentional damage or sabotage	Disqualification from the championship	Without appeal.

Table 13. Penalty schedule.

10.1.1 Score can never go below zero for a task; penalties are deducted from the run total, which is also floored at zero.

10.1.2 Repeated minor infractions may be escalated by the Head Referee.

10.1.3 Teams are responsible for the conduct of every team member and supporter.

11. TEAM CONDUCT AND FAIR PLAY

11.1 Conduct Rules

11.1.1 Teams are made up of interns of the programme and have an appointed captain who is the only point of contact with officials.

11.1.2 The rover must be designed and built by the team. External help in the form of off-the-shelf components is allowed; fully purchased rovers are not.

11.1.3 Sharing knowledge between teams is encouraged after the event. During the event teams may not interfere with other teams' rovers, equipment or communication.

11.1.4 Teams must be respectful to referees, volunteers, and other teams. Abuse leads to disqualification.

11.1.5 Cheating, including tampering with scoring elements or communication, leads to disqualification.

12. RANKING AND AWARDS

12.1 Final Ranking

12.1.1 The final ranking is determined by the total score: the sum of the best Stage 1 run, the best Stage 2 run and the sum of the Stage 3 runs, plus the Dashboard Score and the Build Quality Score (the Software & Autonomous Rulebook).

12.1.2 Ties are broken in the following order: (1) higher autonomous points, (2) higher class, (3) fewer resets and interventions, (4) fastest time to the last scored task, (5) a tie-break run decided by the Head Referee.

12.2 Awards

Award	Criterion
Champion, Runner-up, Third place	Highest final ranking.
Best in Mobility	Highest mobility points over Stage 1 and Stage 2.
Best in Dexterity	Highest dexterity and sensor crate points.
Best in Autonomy	Highest autonomous points.
Best Dashboard	Highest Dashboard Score.
Best Engineering	Design, build quality (including the Build Quality Score), documentation and milestone progress as judged by the Technical Committee.

Table 14. Awards.

13. PROTESTS AND RULE INTERPRETATION

13.1 Protest Procedure

13.1.1 A protest must be submitted in writing by the team captain to the Head Referee within 15 minutes of the end of the run concerned.

13.1.2 The Head Referee decides the protest after consulting the referee of the run, the Technical Committee and, if necessary, video evidence. The decision is final.

13.1.3 Questions about the interpretation of the rules before the event shall be sent to the organizing committee, which publishes the answers to all teams.

13.1.4 The organizing committee may amend the rules, with notice, up to the technical briefing. Changes after that point are made only for safety reasons.

14. REVISION HISTORY

Version	Date	Changes
1.0	October 2026	First edition of the URRT Championship rulebook.
1.1	October 2026	Removed arm, gripper, mass limit and handle requirement; added arena specification and walls; added four team milestones; four-team format.
1.2	October 2026	Added Dashboard Challenge: requirements, interface contract, marking scheme, review and marking sheet; dashboard added to milestones, inspection, ranking and awards.
1.3	October 2026	Added mechanical and electrical design requirements (CAD, 3D-printed chassis, soldered circuits, XT/T connectors, wire colours, schematics) and the Build Quality Score with its M4 Build Review.

Table 15. Revision history.